Vision Aided Navigation - Exercise 5

Student: Shay Morad

GitHub Repository: https://github.com/ShayMorad/3D-Computer-Vision-SLAM

(Note: Code is located on the main branch. Collaborator invites have been sent to course teachers).

Part 5.1: Global Tracking Statistics

To transition from the deterministic tracking pipeline of previous exercises into the probabilistic factor graph backend required for Bundle Adjustment, the coordinate systems must be mathematically aligned. A valid track spanning 10 frames was isolated from the TrackingDB. The global 3D coordinate of the tracked feature was triangulated using a gtsam.StereoCamera initialized at the last frame of the track. This coordinate was subsequently projected backward through the trajectory to measure accumulated tracking drift.

- ***Question:*** *For extrinsic camera matrix [R|t], what would be the transformation from the camera's coordinate system to the global coordinate system?*
- ***Answer:*** The standard OpenCV Extrinsic matrix, calculated via PnP or extracted from the KITTI Ground Truth, maps points from the Global World coordinate system to the local Camera coordinate system:

  $$X_c = R \cdot X_w + t$$

  GTSAM's Pose3 object explicitly requires the inverse transformation, mapping from the Camera coordinate system back to the Global World coordinate system. To derive this, X_w is isolated:

  $$X_w = R^T \cdot X_c - R^T \cdot t$$

  Therefore, the transformation parameters defining the camera's pose in global coordinates are:

  $$R_{gtsam} = R^T$$
  $$t_{gtsam} = -R^T t$$

Visual Analysis: Reprojection Error vs. Factor Error

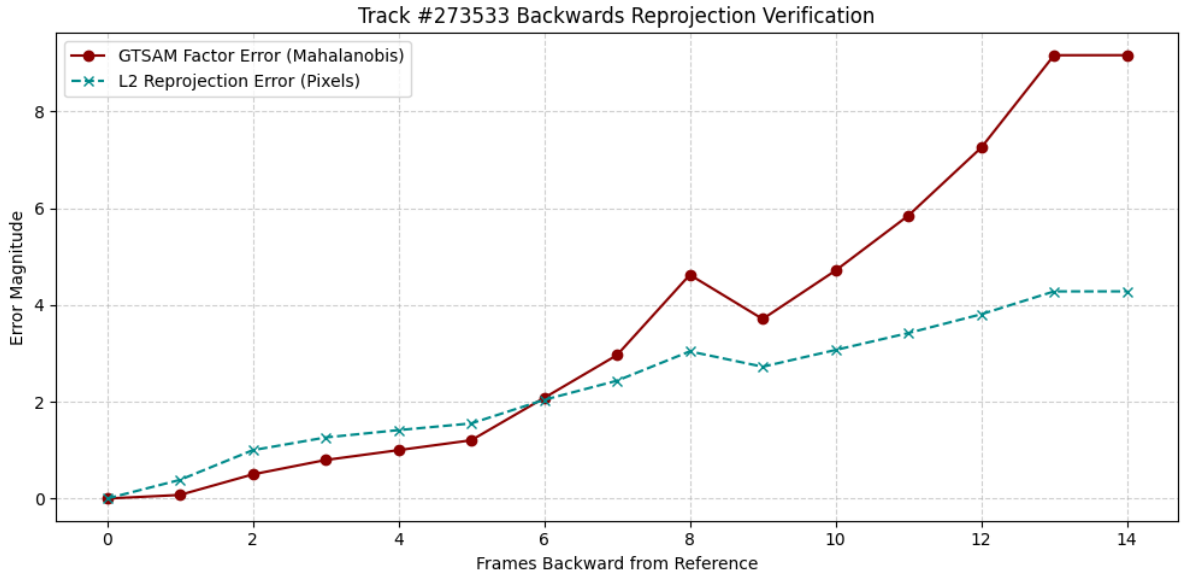


Figure 1: Comparison of L_2 Reprojection Error and GTSAM Factor Error over tracking distance.

Code Reference: ex5.py, Line 77.

- ***Question:*** *What covariance matrix did you use?*
- ***Answer:*** An isotropic Gaussian noise model was applied to the stereo pixel measurements, assuming an independent standard deviation of 1.0 pixel for each coordinate. Using GTSAM's Python wrapper, this is instantiated as gtsam.noiseModel.Isotropic.Sigma(3, 1.0). The resulting 3×3 covariance matrix Σ is the identity matrix I:

$$\Sigma = \begin{bmatrix} 1.0 & 0 & 0 \\ 0 & 1.0 & 0 \\ 0 & 0 & 1.0 \end{bmatrix}$$

The choice of $\sigma = 1.0$ is not arbitrary; it mathematically aligns the probabilistic backend with the deterministic front-end developed in previous exercises. In our front-end architecture, the geometric and RANSAC filters were strictly tuned to reject feature matches with pixel deviations exceeding 2.0 pixels. In a normal distribution, approximately 95% of the data falls within 2σ. By setting $\sigma = 1.0$, we instruct the GTSAM optimizer to expect 95% of valid measurements to naturally fall within this exact 2.0-pixel tolerance, perfectly bridging our front-end outlier rejection with our backend optimization expectations.

- ***Question:*** *What is the factor error as a function of the reprojection error?*
- ***Answer:*** The standard reprojection error (L_2 norm) is the Euclidean distance in pixels. GTSAM optimizes the factor graph by minimizing the *whitened* squared error (the Mahalanobis distance). The unweighted factor error formula evaluated by the

GenericStereoFactor3D is $\frac{1}{2}e^T\Sigma^{-1}$e .

Because our covariance matrix Σ is the identity matrix I, the formula simplifies directly to half the squared Euclidean distance:

$$E_{factor} = \frac{1}{2}E_{proj}^2$$

Part 5.2: Relative Transformation Theory

- ***Question****: Two cameras A, B have extrinsic camera matrices ($[R_A|t_A]$, $[R_B|t_B]$ respectively) in some common coordinate system. Express transformation $T_{A->B}$ from the coordinates of camera A to the coordinates of camera B using R_A, R, t_A, t_B. What would be the extrinsic camera matrix of B in the coordinate system of A?*
- ***Answer:*** To map a 3D point from the local coordinate system of Camera A to the local coordinate system of Camera B, the point must first be transformed from A into the common global World system, and then from the World system into B.
    - **From A to World:** Using the inverted transformation derived in Q5.1:
    $$X_w = R_A^T X_A - R_A^T t_A$$
    - **From World to B:** Applying the standard extrinsic matrix of B:
    $$X_B = R_B X_w + t_B$$
    - **Substitution:** Substituting X_w into the second equation yields:
    $$X_B = R_B(R_A^T X_A - R_A^T t_A) + t_B$$
    $$X_B = (R_B R_B^T)X_A + (t_B - R_B R_A^T t_A)$$
    - Therefore, the relative extrinsic camera matrix $T_{A->B}$, which defines Camera B in the coordinate system of Camera A, is:
    $$R_{A\to B} = R_B R_A^T$$
    $$t_{A\to B} = t_B - R_B R_A^T t_A$$

Part 5.3: Bundle Adjustment Window

To contain computational cost, Bundle Adjustment is performed locally on short segments of consecutive frames. The first bundle window was established to validate the factor graph construction and optimization mechanics.

Keyframe Criterion: Rather than relying on a naive temporal interval (e.g., every 10 frames), a dynamic Distance-Travelled Criterion was implemented to guarantee sufficient parallax for stereo triangulation. A new keyframe is established only when the vehicle translates a minimum Euclidean distance of 5.0 meters from the previous keyframe. A secondary failsafe rule forces a keyframe if 15 frames elapse without reaching the distance threshold, ensuring the bundle window never exceeds the computational bounds requested by the exercise (5-20 frames). The first Bundle window consists of Keyframe 1 (Frame 0) and Keyframe 2, capturing all intermediate frames containing the relevant tracking data.

Graph Formulation & Constraints:

To prevent the optimization from diverging due to Gauge Freedom (ill-formed infinite coordinate solutions), a PriorFactorPose3 with a minimal covariance (1×10^{-6}) was anchored to Frame 0. Initial trajectory estimates were provided by the PnP matrices from Exercise 3, and initial 3D landmark values were triangulated from their final observation within the bundle.

Optimization Metrics:

- **Total Factors in Graph:** 6,317
- **Total Factor Graph Error:**
  - Before Optimization: 25,395.63
  - After Optimization: 959.22
- **Average Factor Error:**
  - Before Optimization: 4.02
  - After Optimization: 0.15

```
--- Q 5.3: Bundle Adjustment Window ---
Dynamic Keyframe Criterion: Distance > 5.0m or Frames >= 15
Bundle Window Bounds: Frames 0 to 7

--- BEFORE OPTIMIZATION ---
Total Factors: 6317
Graph Error: 25395.63 | Average Error: 4.02

Max Initial Error Factor: Frame c=0, Track q=149
Initial Factor Error: 90.43
Distance from Measurements -> Left: 9.95px, Right: 9.38px

--- AFTER OPTIMIZATION ---
Total Graph Error: 959.22 | Average Error: 0.15

Re-evaluating Factor: Frame c=0, Track q=149
Corrected Target Factor Error: 0.85
Distance from Measurements -> Left: 1.28px, Right: 1.30px

FINISHED: --- SECTION 5.3 ---
```

Worst Factor Analysis

To verify the geometric integrity of the Levenberg-Marquardt optimization, the projection factor with the largest initial error was isolated. This factor belonged to Frame $c = 0$ and 3D point $q = 149$.

- Initial State (Before BA): The initial factor error was 90.43 . Projecting the initialized coordinate onto the images resulted in a severe Euclidean distance error of 9.95 pixels from the Left measurement and 9.38 pixels from the Right measurement.

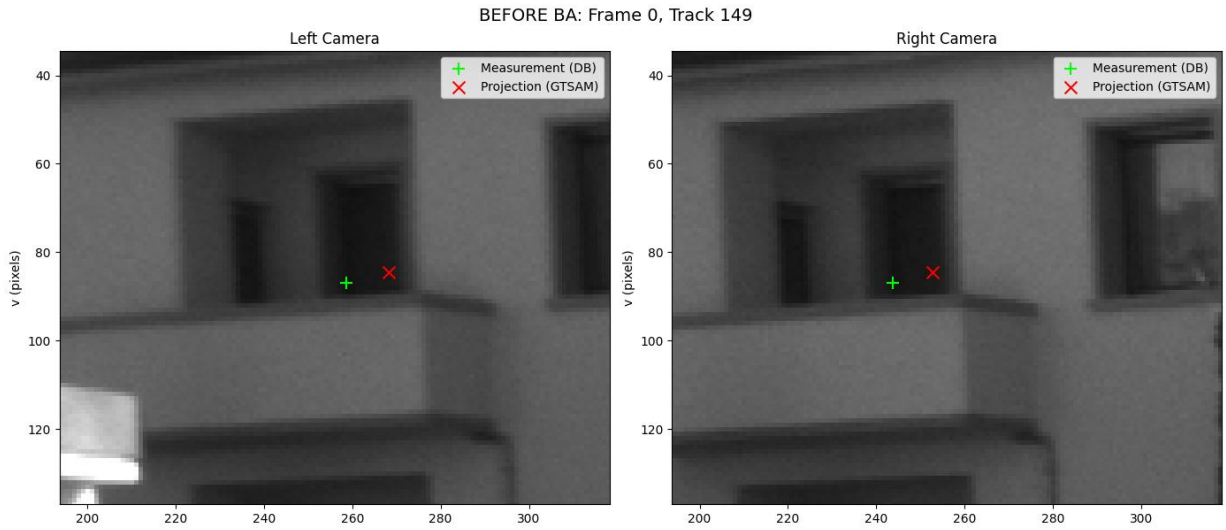


Figure 2: Before optimization: Ground Truth measurement (Green) vs GTSAM Initial Projection (Red).

Code Reference: ex5.py, Line 118.

- **Optimized State (After BA):** Following the global graph optimization, the factor error collapsed to just 0.85. The optimized projection successfully converged on the visual measurement, leaving a negligible sub-pixel error distance of 1.28 pixels on the Left and 1.30 pixels on the Right.

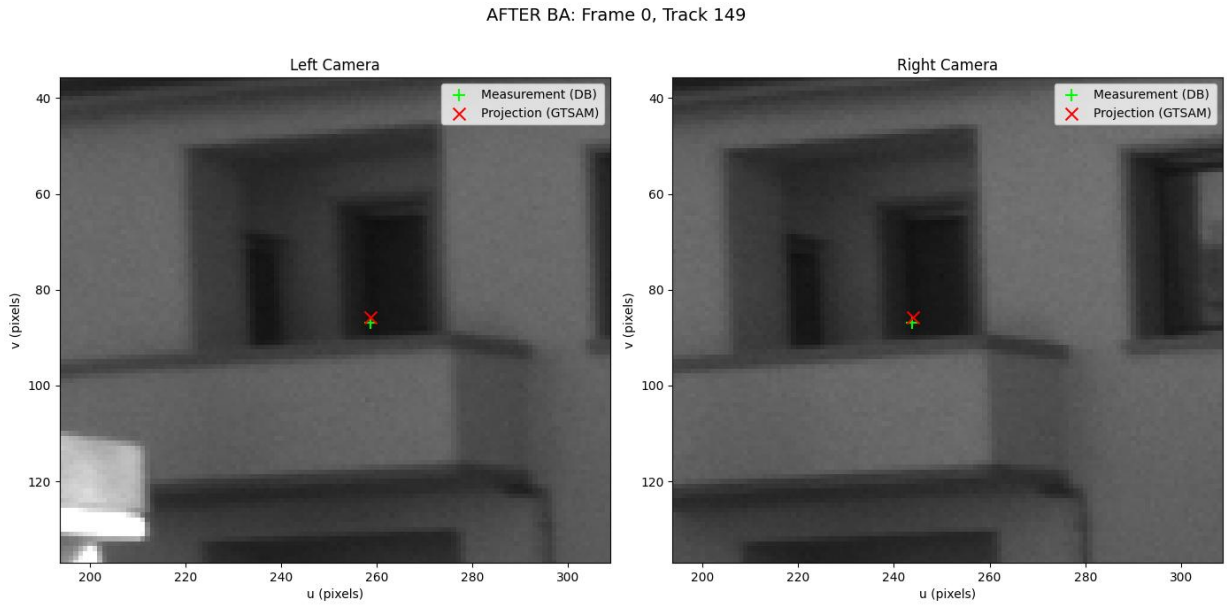


Figure 3: After optimization: The GTSAM Projection successfully snaps to the actual geometric measurement.

Code Reference: ex5.py, Line 136.

Local Bundle Trajectory Visualization

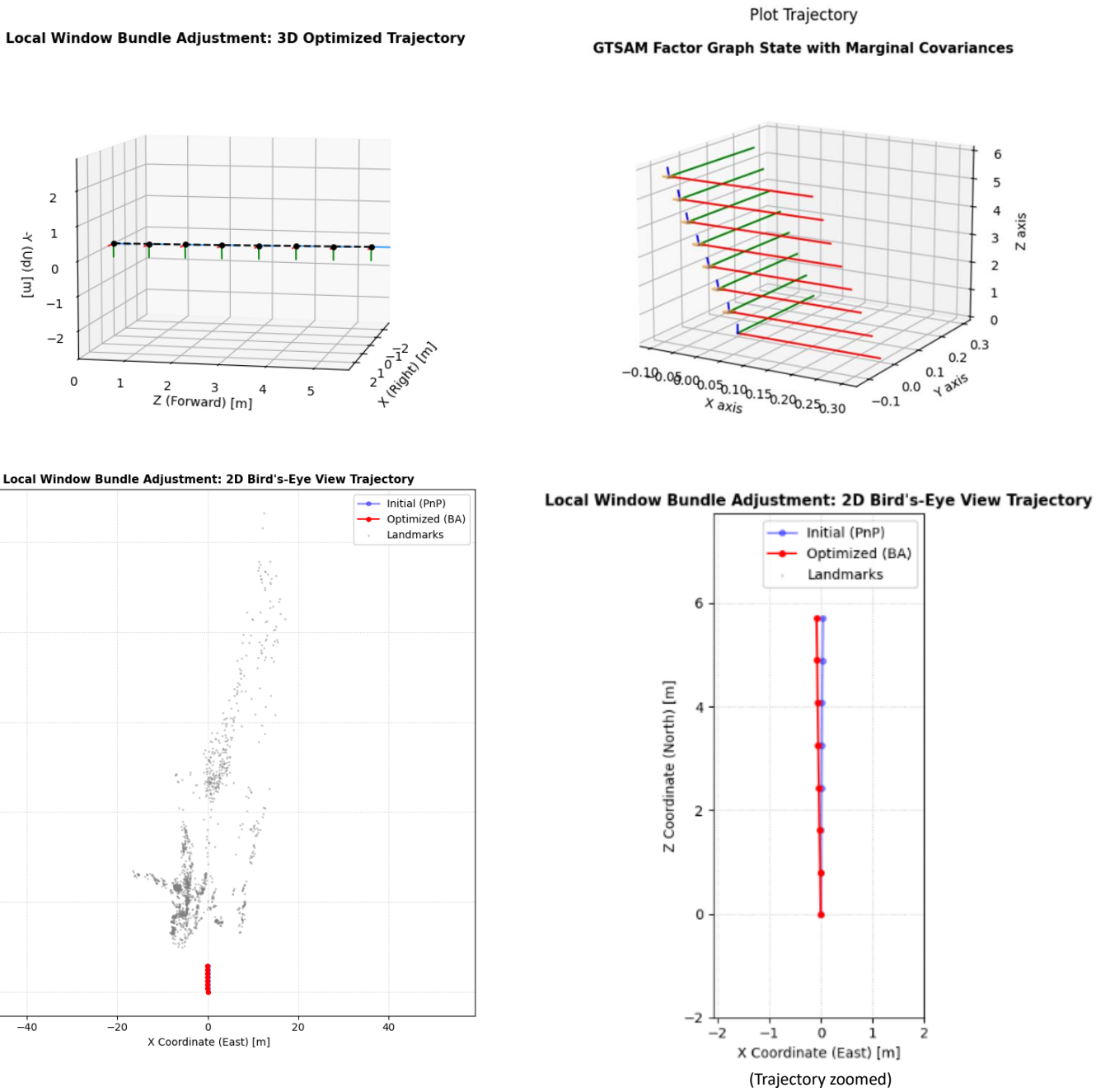


Figure 4: Trajectory of Bundle 1. The 3D plot visualizes the refined camera poses, while the 2D plane demonstrates the optimized trajectory mapped against the initial PnP estimates alongside the reconstructed 3D landmark point cloud.

Code Reference: ex5.py, Line 141.

Part 5.4: Global Sliding Window & Map Chaining

To generate the final visual odometry, the local bundle adjustment was transitioned into a global sliding window pipeline. A dynamic distance-travelled criterion was utilized to select keyframes ensuring sufficient baseline translation ($\approx 5.0m$) between optimized windows.

To ensure numerical stability and avoid floating-point precision loss over the trajectory, each bundle was computed relatively. The initial values were transformed into the local coordinate system of the bundle's first frame, and a PriorFactorPose3 was used to strictly anchor that first frame to the local origin (0, 0, 0). Following the Levenberg-Marquardt optimization of each window, the refined relative pose between the keyframes $T_{kf-1 \to kf}$ was extracted and chained into the global trajectory sequence using standard transformation matrix composition ($T_{global} = T_0 \cdot T_1 \dots T_n$).

Theoretical Analysis: The Last Bundle

```
--- LAST BUNDLE THEORETICAL ANALYSIS ---
Optimized Anchor Position (X, Y, Z):
[-2.20275965e-23 -4.51301170e-24  1.18654976e-22]
Anchoring Factor Final Error: 9.563491989802383e-33
```

- *Because the optimization was executed in the relative coordinate system of the first frame, its optimized position remains perfectly locked at the origin. The terminal output confirms this, with the position returning as computational zero: X: -2.20275965e-23, Y: -4.51301170e-24, Z: 1.18654976e-22*
- ***Question:** Print the anchoring factor final error. Why is that the error?*
- ***Answer:** The anchoring factor final error evaluated to 9.563491989802383e-33 (effectively 0.0). This error is zero because the PriorFactorPose3 acts as a hard constraint demanding the first camera remain at the origin (0,0,0). Because all other factors in this local graph are purely relative (stereo pixel measurements mapping cameras to local 3D landmarks), the graph contains no conflicting global measurements. The optimizer can resolve the graph entirely through relative spatial adjustments, satisfying the prior factor constraint perfectly without introducing any tension or residual penalty.*

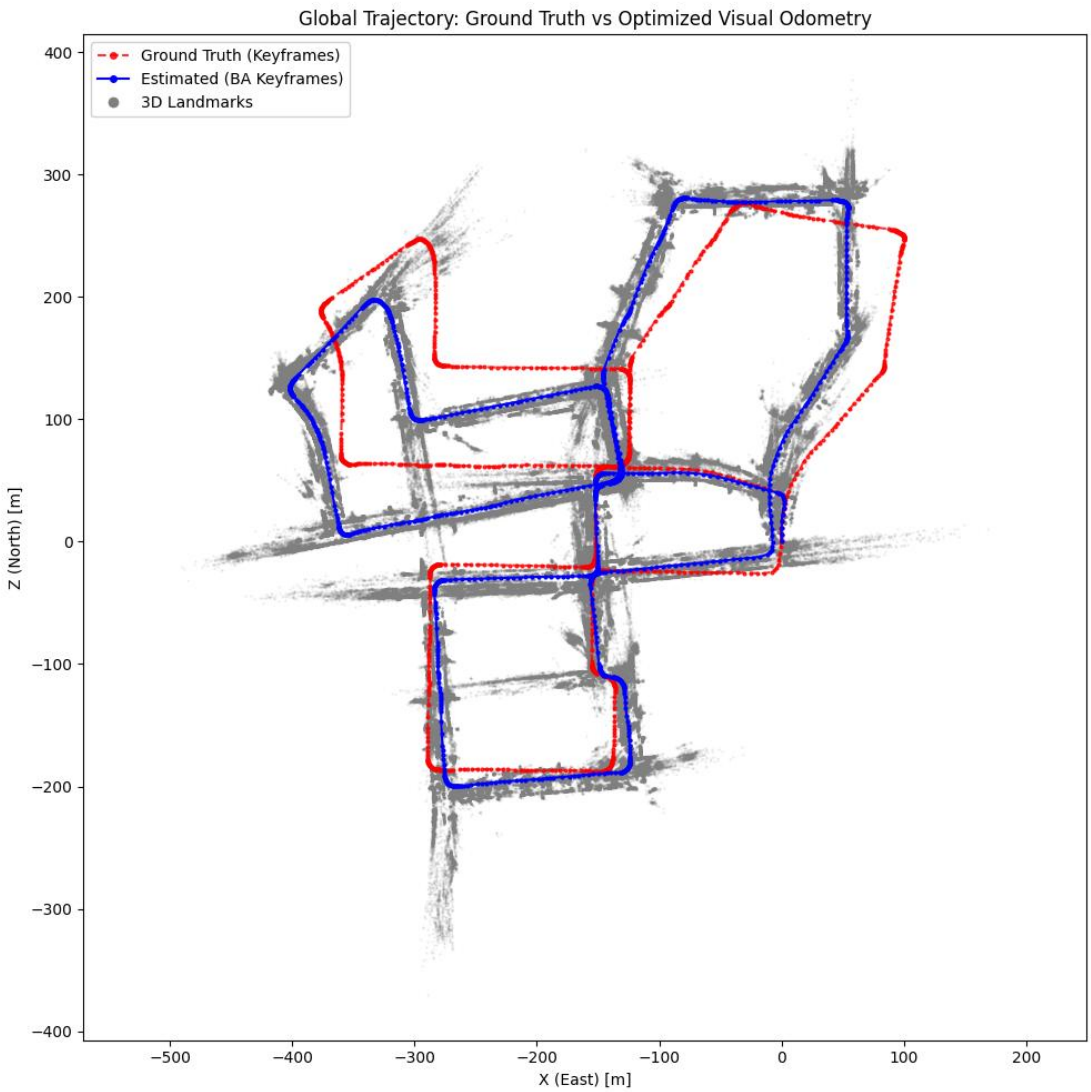


Figure 5: 2D Bird's-Eye View combining the Estimated Visual Odometry (Blue) representing the left-camera keyframes, the Ground Truth poses (Red Dashed), and the triangulated 3D map landmarks (Gray).

Code Reference: ex5.py, Line 214.

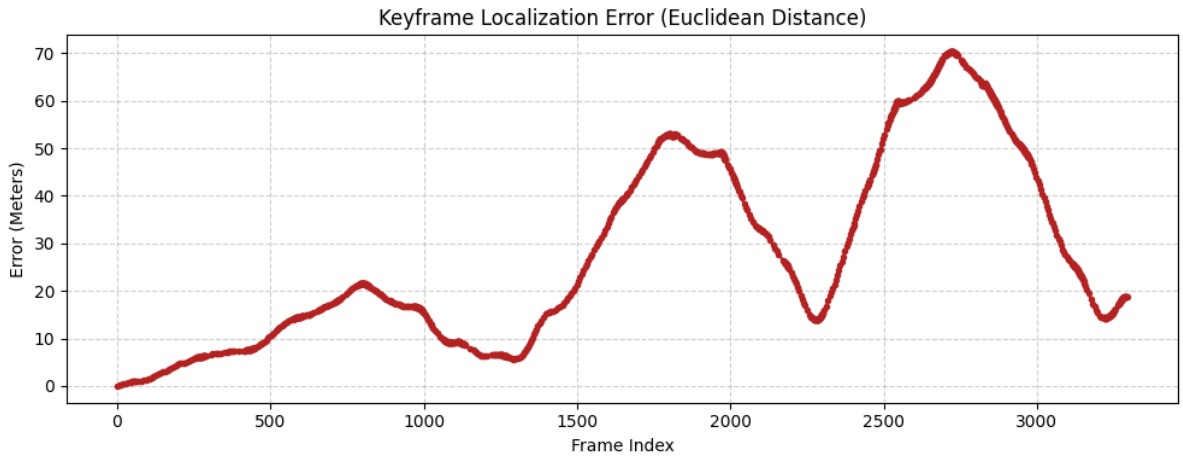


Figure 6: Euclidean Localization Error tracking the progressive physical drift (in meters) between the estimated visual odometry and the ground truth across 3,300 frames.

Code Reference: ex5.py, Line 215.

Trajectory Analysis: The plotted Visual Odometry trajectory in Figure 5 successfully reconstructs the macro-structure of the KITTI sequence, preserving all major loops, turns, and orthogonal street grids. However, a progressive divergence between the Estimated Trajectory and the Ground Truth is visible as the sequence advances.

This divergence, quantified in Figure 6, is a textbook manifestation of **Odometry Drift (Integration Error)**. The Levenberg-Marquardt optimizer was executed over local sliding windows. While this Local Bundle Adjustment highly refines the relative transformations and eliminates local tracking jitter, the global system is fundamentally operating via dead reckoning. Microscopic sub-pixel uncertainties in stereo disparity and feature localization manifest as marginal rotational and translational errors within each local bundle.

Because the global trajectory is generated by chaining these relative transformations sequentially, these marginal angular errors compound. An imperceptible fraction of a degree in rotation error at Frame 500 amplifies into meters of translational error by Frame 3000. Without global constraints this localized drift inevitably translates into the growing, fluctuating Euclidean localization error demonstrated in the final metrics.